Logging data you can trust

IoT · Level 3 · Lesson 1



Today we will

- Plan a data log: what to measure, where, how often, with a time and unit on every reading.
- Check a sensor against a reference before trusting it.
- Clean glitches openly, and use the median when a few bad readings would drag the average.

Data you can trust

- Start with a question
- Time, unit and place on every reading
- Check against a reference
- Clean openly; report the median

A paper log on a rota

1. Warm-up (5 min): Show “85 °C at 7 am”. Could it be true?
2. Plan (8 min): Agree the question: what, where, how often.
3. Check (7 min): Read two thermometers side by side.
4. Log (12 min): Start the rota; practise readings with time and unit.
5. Wrap-up (8 min): Sort sample readings and find the median.

Before class, hang the thermometer in the shade at desk height and check that pupils can read it. Use a dial or digital thermometer with young pupils; if a glass one breaks, an adult clears it up. If you use a sensor board, run it on batteries only.

Let’s discuss

1. Where else at school could a glitch mislead a decision?
2. When should a surprising reading stay in the data?
3. Who should be able to check how the data was cleaned?

Exit ticket

- Readings: 30, 28, 85, 31, 29 °C. What is the median, and what goes in the cleaning note?